Report — Task 2: Analysis Report

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Architectural Changes

```
class GreenNet(nn.Module):
    """
    Two Conv-ReLU-MaxPool blocks, followed by Average Pooling and a single FC layer.
    """
    def __init__(self, in_channels=3, num_classes=4, **kwargs):
        super().__init__()
        self.conv1 = nn.Conv2d(in_channels, 16, kernel_size=3, padding=1)
        self.bn1 = nn.BatchNorm2d(16)
        self.conv2 = nn.Conv2d(16, 32, kernel_size=3, padding=1)
        self.bn2 = nn.BatchNorm2d(32)
        self.relu = nn.ReLU()
        self.pool = nn.MaxPool2d(2)
        self.gap = nn.AdaptiveAvgPool2d(1)
        self.fc = nn.Linear(32, num_classes)

    def forward(self, x):
        x = self.pool(self.relu(self.conv1(x)))
        x = self.pool(self.relu(self.conv2(x)))
        x = self.gap(x)
        x = x.flatten(1)
        return self.fc(x)
```

Efficiency Verification Matrix

Averaged across all four datasets (cells, chest, lesions, orgs):

Model	Avg. Train Time (s)	Avg. Peak Train Memory (MB)	Avg. Inference Latency (ms/sample)	Avg. Accuracy
GreenNet	16.66	71.7	0.880	70.95%
AlexNet	46.40	186.3	2.317	84.86%
VGG16	330.60	854.4	6.070	83.06%
ResNet18	649.41	1349.3	8.346	83.46%

GreenNet trains roughly 3x faster than AlexNet, 20x faster than VGG16, and 39x faster than ResNet18, while using **2.6x less memory than AlexNet, 12x less than VGG16, and 19x less than ResNet18**. Inference latency follows the same pattern: GreenNet responds in under a millisecond per sample, compared to over 8ms for ResNet18 — a roughly 9x difference.

Accuracy Trade-Off

Dataset	GreenNet Accuracy	Best Baseline Accuracy	Gap
cells	69.07%	95.57% (VGG16)	-26.50
chest	75.96%	89.58% (ResNet18)	-13.62
lesions	67.13%	73.34% (AlexNet)	-6.21
orgs	71.63%	91.77% (ResNet18)	-20.14